

Genetic Analysis of Disease in Diverse Individuals

LA's Best

6/15/2026

Nicholas Mancuso, PhD

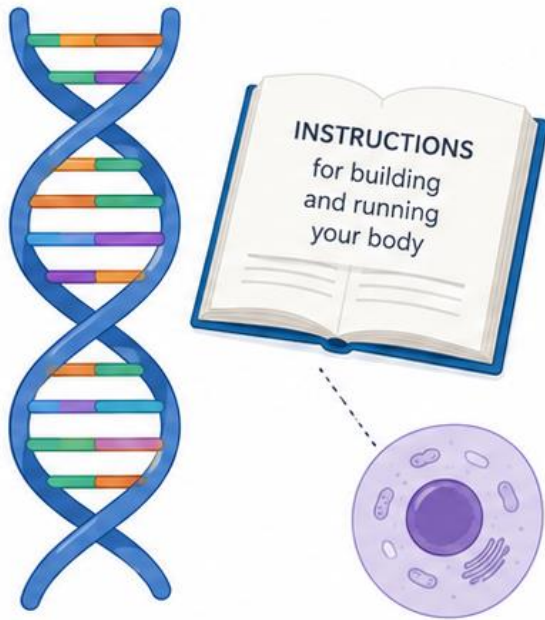
Center for Genetic Epidemiology

Keck School of Medicine, USC

Keck School of
Medicine of **USC**

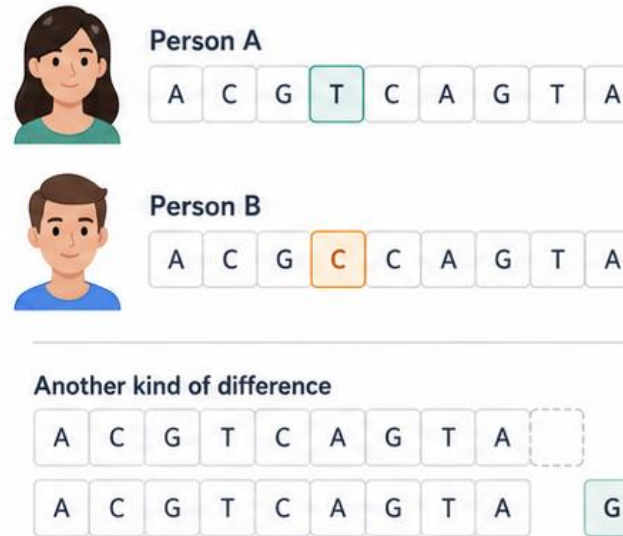
DNA and genetic variation

1. DNA stores biological instructions



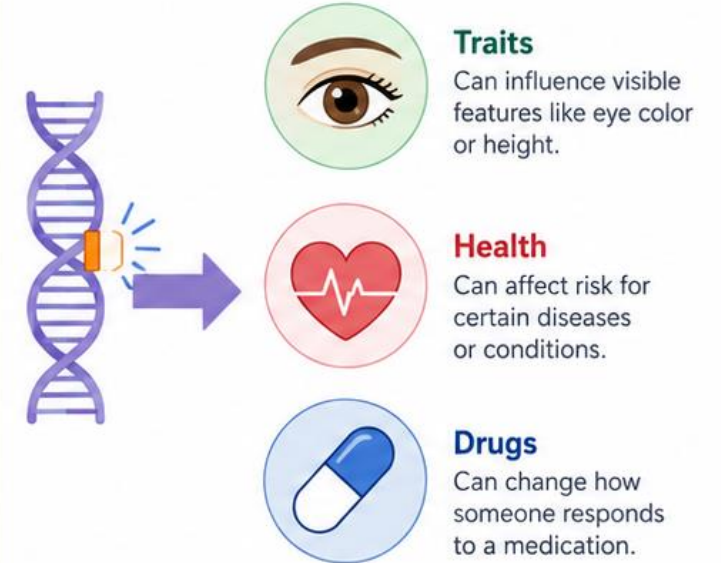
DNA is a long molecule made of A, C, G, and T.
It carries instructions used by cells.

2. Genetic variation = small differences in the code



Most DNA is shared across people, but some letters or segments differ. These differences are called variants.

3. Some variants affect traits, health, or drugs

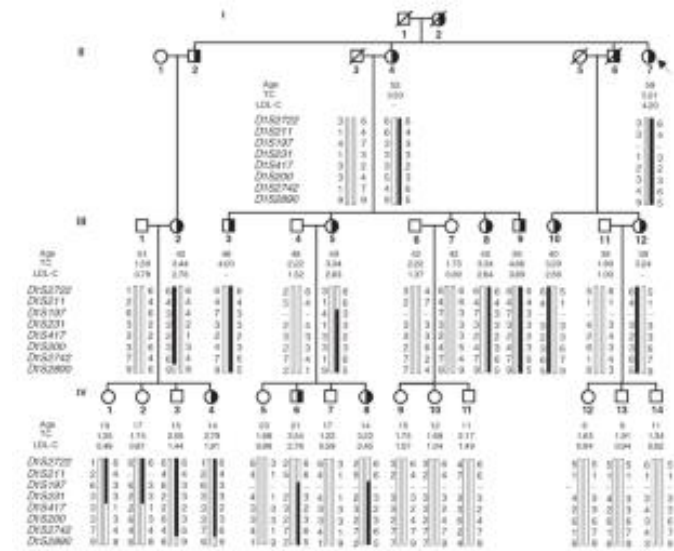


Some variants can influence visible traits, disease risk, or how someone responds to a medication. Most variants are harmless.

Mutations in *PCSK9* cause autosomal dominant hypercholesterolemia

Marianne Abifadel, Mathilde Varret, Jean-Pierre Rabès, Delphine Allard, Khadija Ouguerram, Martine Devillers, Corinne Cruaud, Suzanne Benjannet, Louise Wickham, Danièle Erlich, Aurélie Derré, Ludovic Villéger, Michel Farnier, Isabel Beucler, Eric Bruckert, Jean Chambaz, Bernard Chanu, Jean-Michel Lecerf, Gerald Luc, Philippe Moulin, Jean Weissenbach, Annick Prat, Michel Krempf, Claudine Junien, ... Catherine Boileau  + Show authors

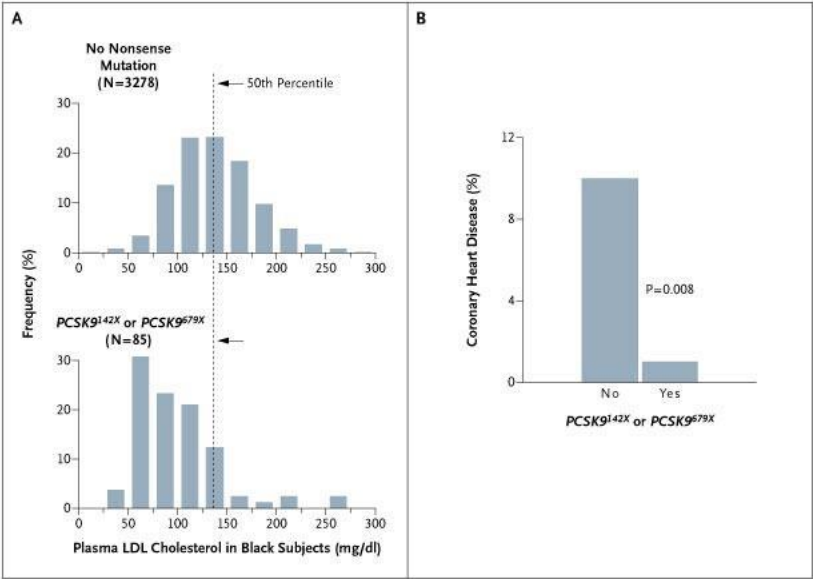
Nature Genetics **34**, 154–156 (2003) | [Cite this article](#)



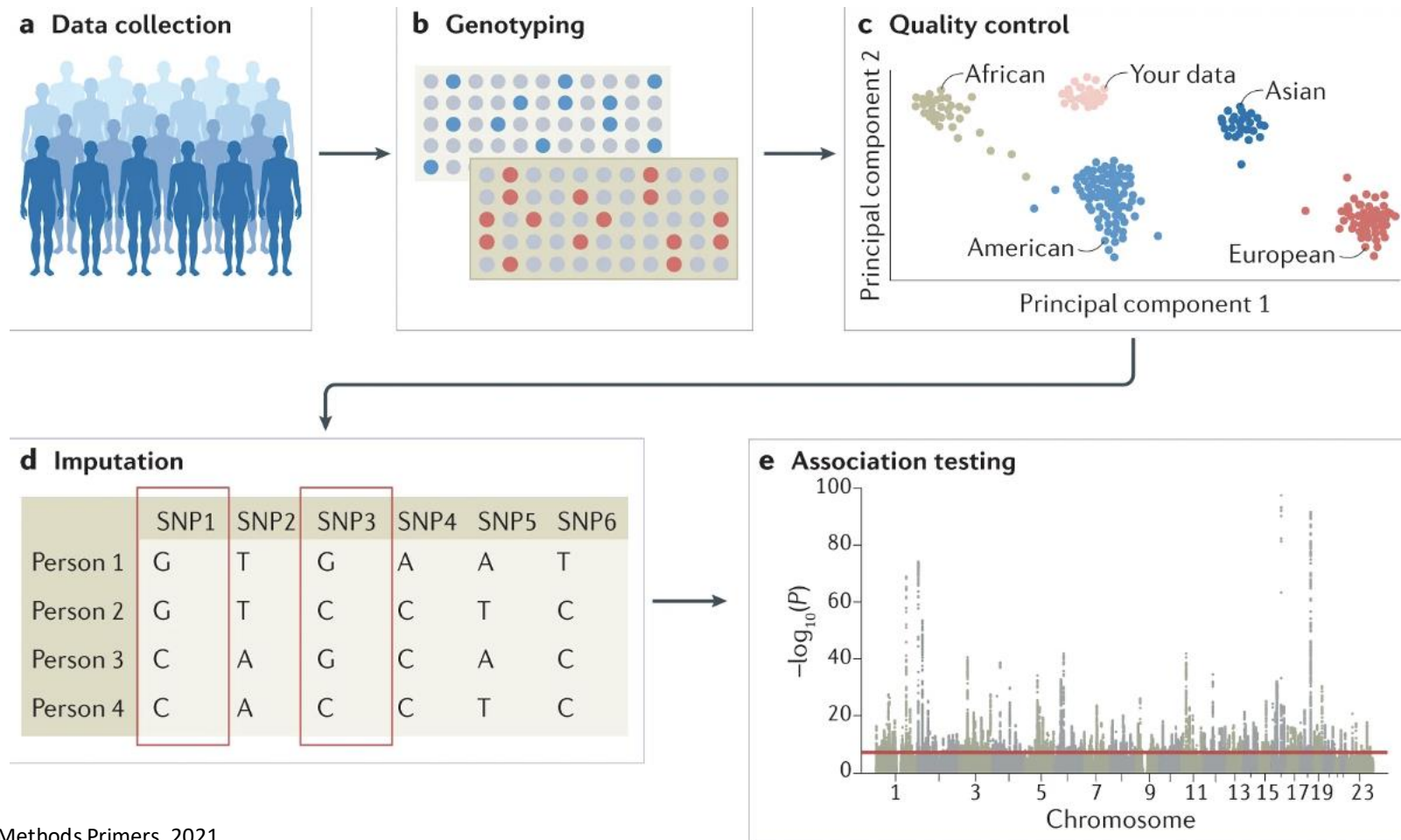
Sequence Variations in *PCSK9*, Low LDL, and Protection against Coronary Heart Disease

Authors: Jonathan C. Cohen, Ph.D., Eric Boerwinkle, Ph.D., Thomas H. Mosley, Jr., Ph.D., and Helen H. Hobbs, M.D. [Author Info & Affiliations](#)

Published March 23, 2006 | *N Engl J Med* 2006;354:1264-1272 | DOI: 10.1056/NEJMoa054013 | **VOL. 354 N1**
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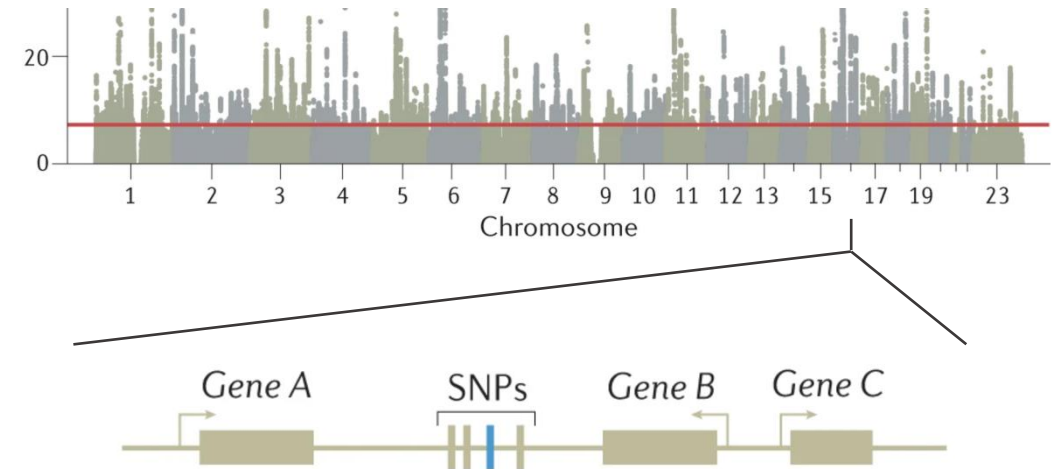


Genome-wide Association Studies (GWASs) identify *genetic variants* associated with *complex traits*

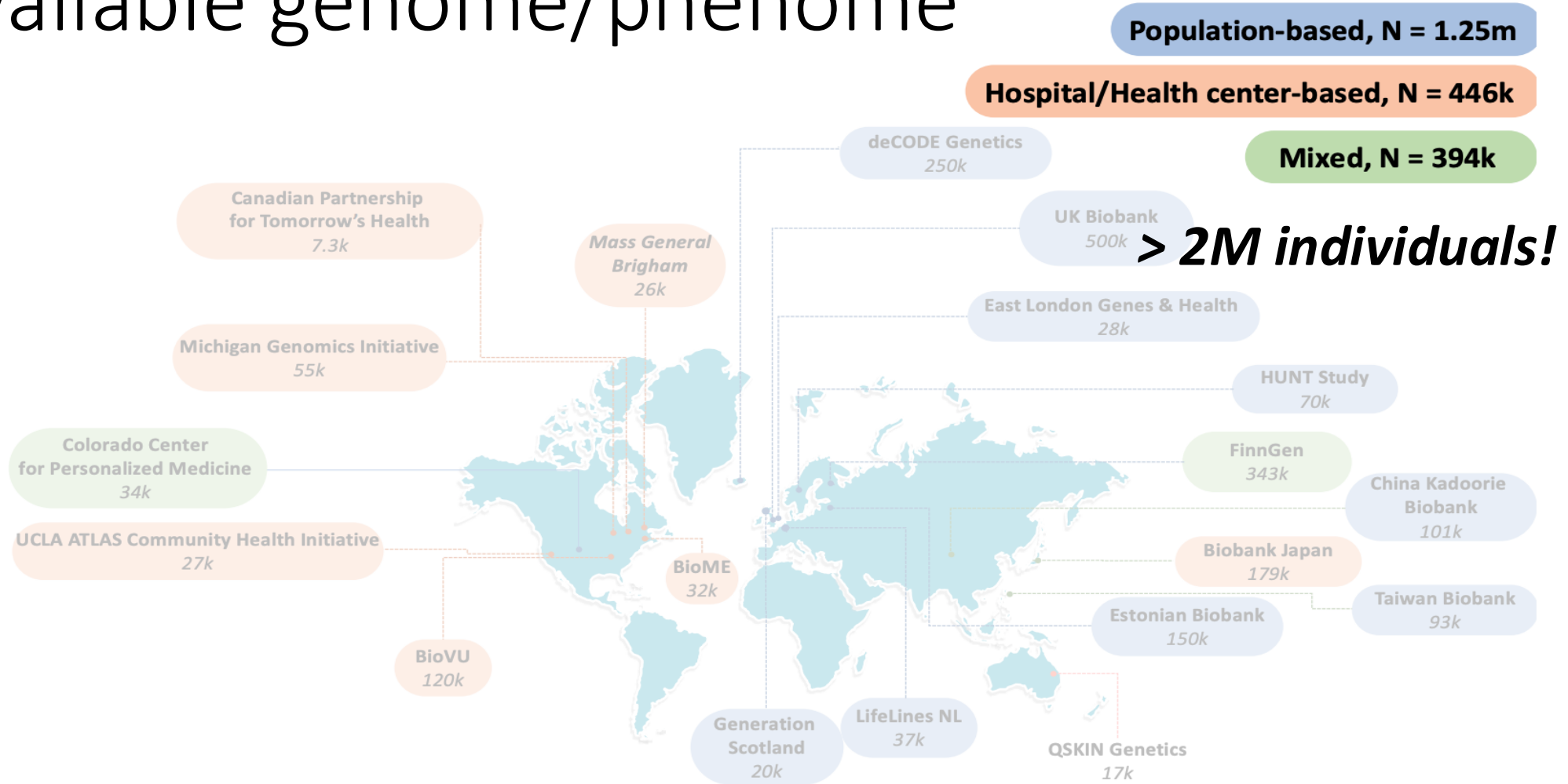


GWAS does not provide *causal mechanisms*

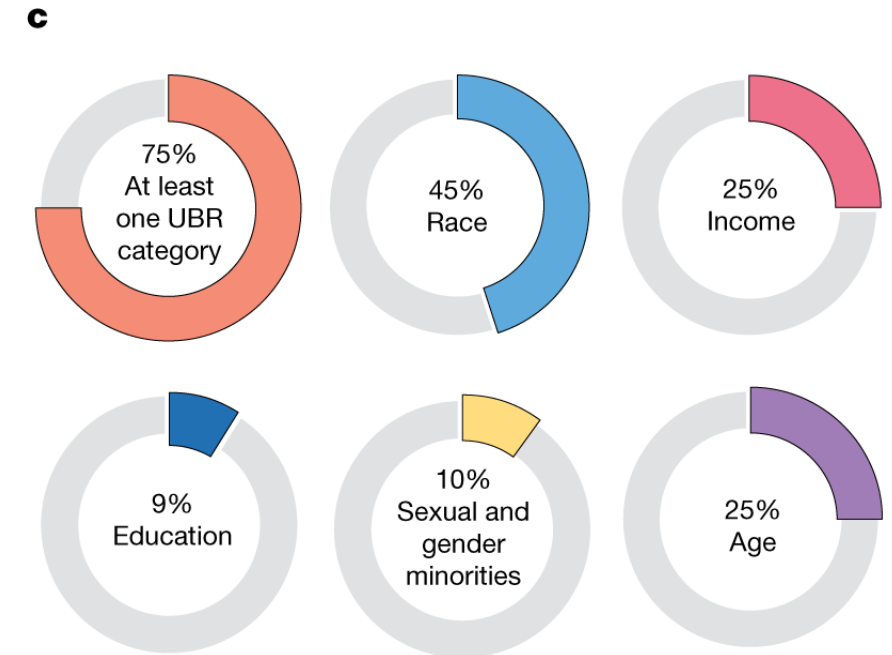
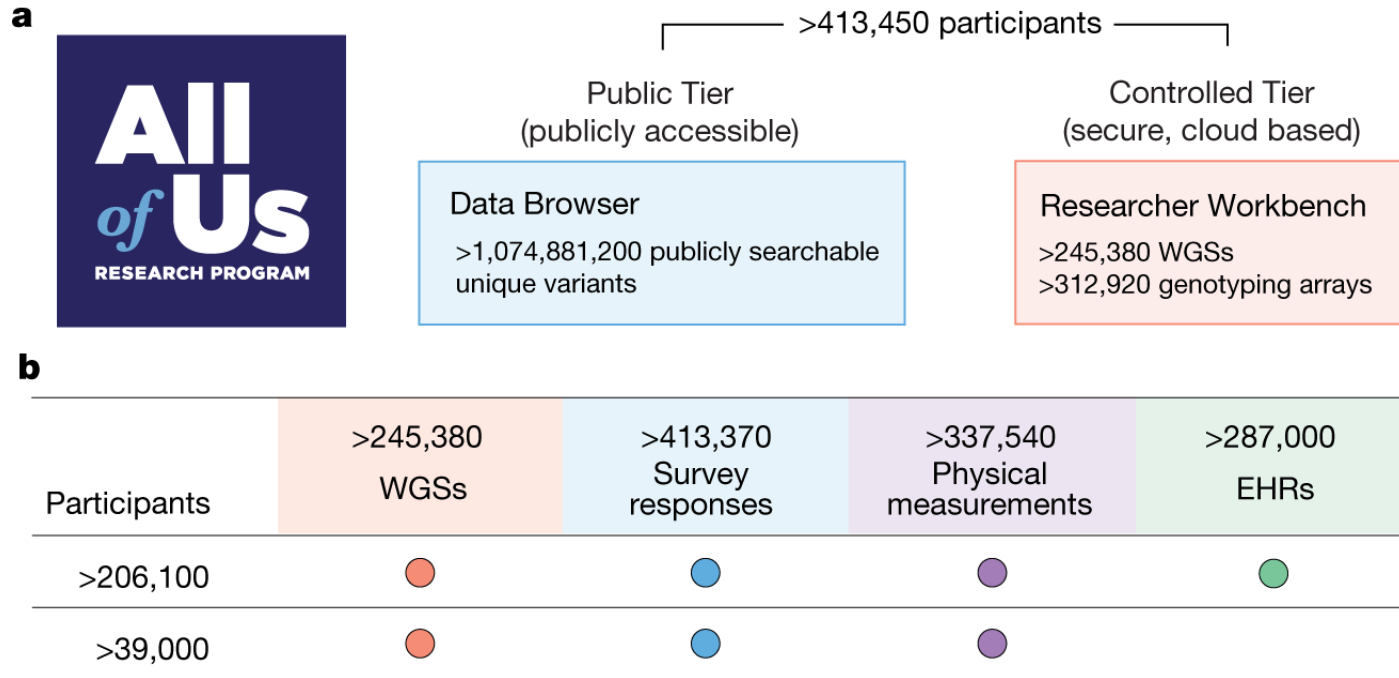
- GWAS provides regions **associated** with disease risk
- What are the biological mechanisms driving disease risk?
- ***Functional architectures***: a collection of functional mechanisms
- Most associations are non-coding
 - Non-coding SNPs are enriched with **regulatory functions on molecular traits**



Global biobanks reflect *massive* scale of available genome/phenome



All of US Biobank



LA's Best Project: All of Us Analysis

- Project Aims:

- Learn to navigate genetic + phenotypic data on AOU portal
- Select phenotype and construct analysis group (non EHR related!)
- Visualize diverse ancestry components using PCA
- Perform GWAS

- Deliverables

- Small presentation/writeup on any potential genetic findings